Version 7.4.8

PVsyst - Simulation report

Grid-Connected System

Project: Qom-1MW

Variant: New simulation variant

Sheds on ground

System power: 1002 kWp

Qom - Iran

Author

PVsyst V7.4.8
VC0, Simulation date:
10/25/26 16:46
with V7.4.8

Project summary

Geographical Site
Qom
Iran

Situation

Latitude	34.65 °N
Longitude	50.90 °E
Altitude	928 m
Time zone	UTC+3.5

Project settings

Albedo	0.20

Weather data
Vādī os Salām
Meteonorm 8.1 (1988-2000), Sat=100% - Synthetic

System summary

Grid-Connected System

Sheds on ground

PV Field Orientation
Fixed plane
Tilt/Azimuth 34 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array

Nb. of modules	1392 units
Pnom total	1002 kWp

Inverters

Nb. of units	6 units
Pnom total	990 kWac
Pnom ratio	1.012

Results summary

Produced Energy	1935790 kWh/year	Specific production	1931 kWh/kWp/year	Perf. Ratio PR	83.11 %

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General parameters

Grid-Connected System

Sheds on ground

PV Field Orientation

Orientation	
Fixed plane	
Tilt/Azimuth	34 / 0 °

Sheds configuration	
Nb. of sheds	58 units
Sizes	
Sheds spacing	8.00 m
Collector width	4.79 m
Ground Cov. Ratio (GCR)	59.9 %
Top inactive band	0.02 m
Bottom inactive band	0.02 m
Shading limit angle	
Limit profile angle	33.8 °

Models used	
Transposition	Perez
Diffuse	Perez, Meteonorm
Circumsolar	separate

Horizon
Free Horizon

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer	Mana Energy
Model	MEP720-132-GB
(Custom parameters definition)	
Unit Nom. Power	720 Wp
Number of PV modules	1392 units
Nominal (STC)	1002 kWp
Modules	58 string x 24 In series
At operating cond. (50°C)	
Pmpp	931 kWp
U mpp	943 V
I mpp	987 A

Inverter

Manufacturer	SMA
Model	Sunny Highpower SHP165-US-21-PEAK3
(Original PVsyst database)	
Unit Nom. Power	165 kWac
Number of inverters	6 units
Total power	990 kWac
Operating voltage	898-1500 V
Pnom ratio (DC:AC)	1.01

Total PV power

Nominal (STC)	1002 kWp
Total	1392 modules
Module area	4321 m²
Cell area	2026 m²

Total inverter power

Total power	990 kWac
Number of inverters	6 units
Pnom ratio	1.01

Array losses

Thermal Loss factor

Module temperature according to irradiance	
Uc (const)	29.0 W/m²K
Uv (wind)	0.0 W/m²K/m/s

DC wiring losses

Global array res.	16 mΩ
Loss Fraction	1.5 % at STC

LID - Light Induced Degradation

Loss Fraction	1.3 %

Module Quality Loss

Loss Fraction	-0.2 %

Module mismatch losses

Loss Fraction	2.0 % at MPP

Strings Mismatch loss

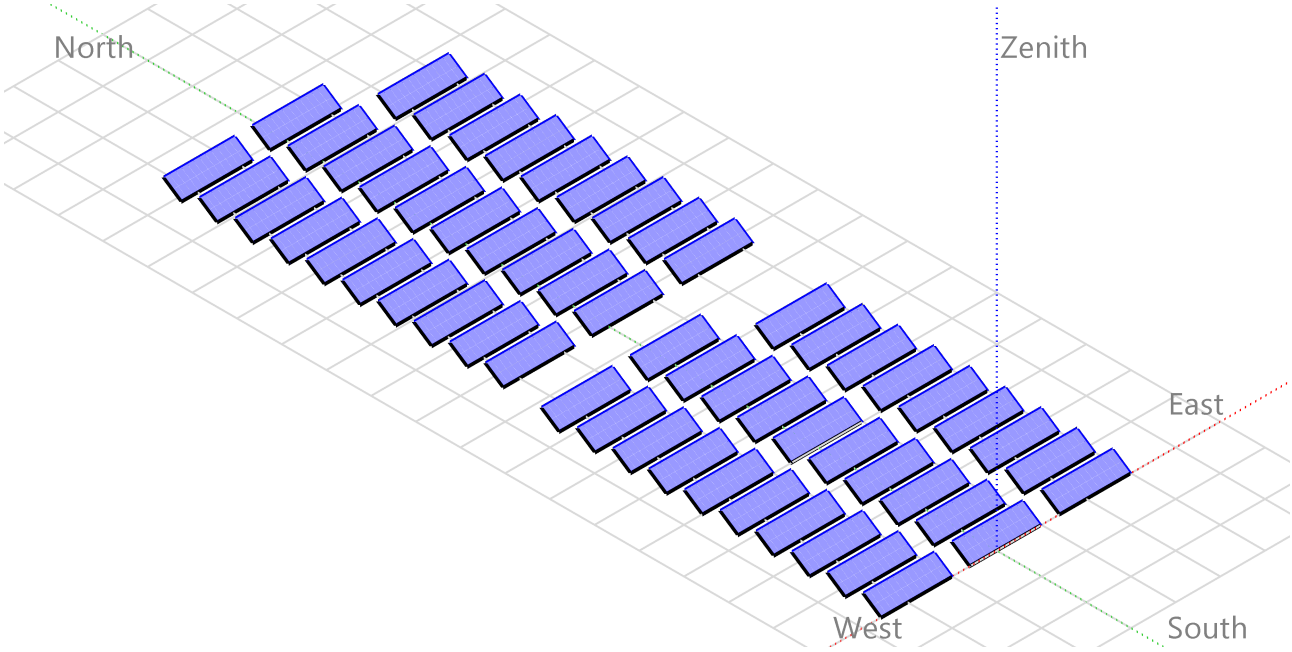
Loss Fraction	0.1 %

IAM loss factor
Incidence effect (IAM): Fresnel smooth glass, n = 1.526

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.998	0.981	0.948	0.862	0.776	0.636	0.403	0.000

Near shadings parameter

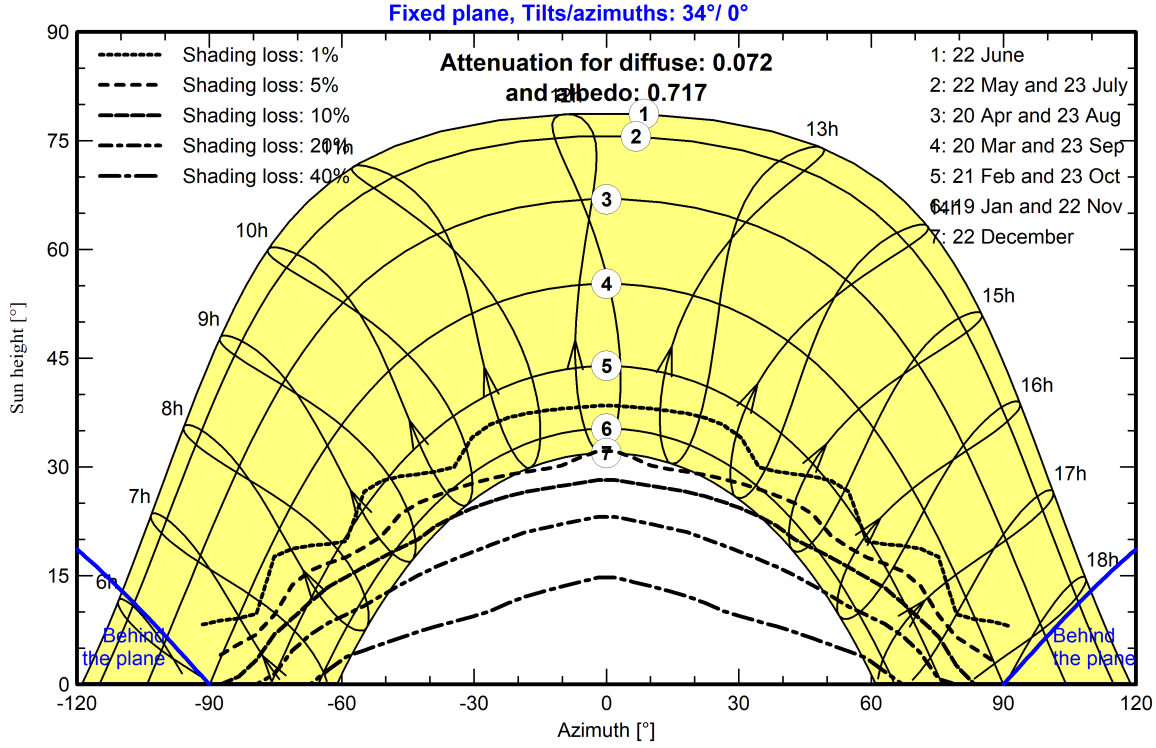
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 34° / 0°

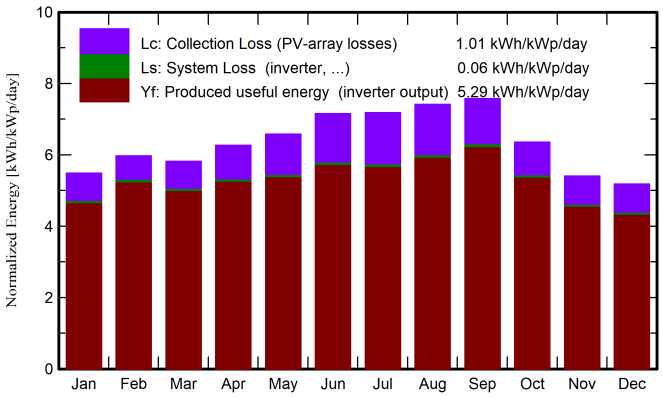


Main results

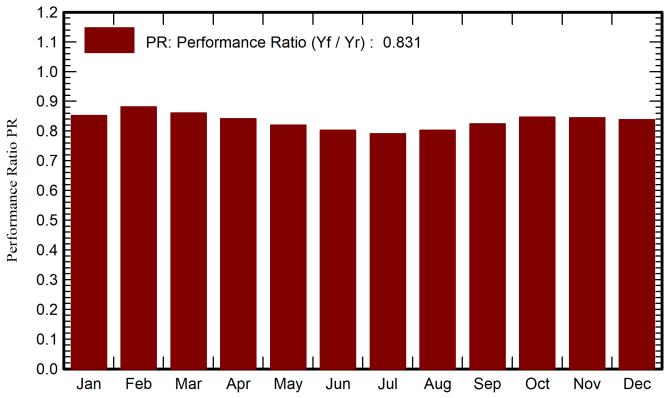
System Production

Produced Energy	1935790 kWh/year	Specific production	1931 kWh/kWp/year
		Perf. Ratio PR	83.11 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

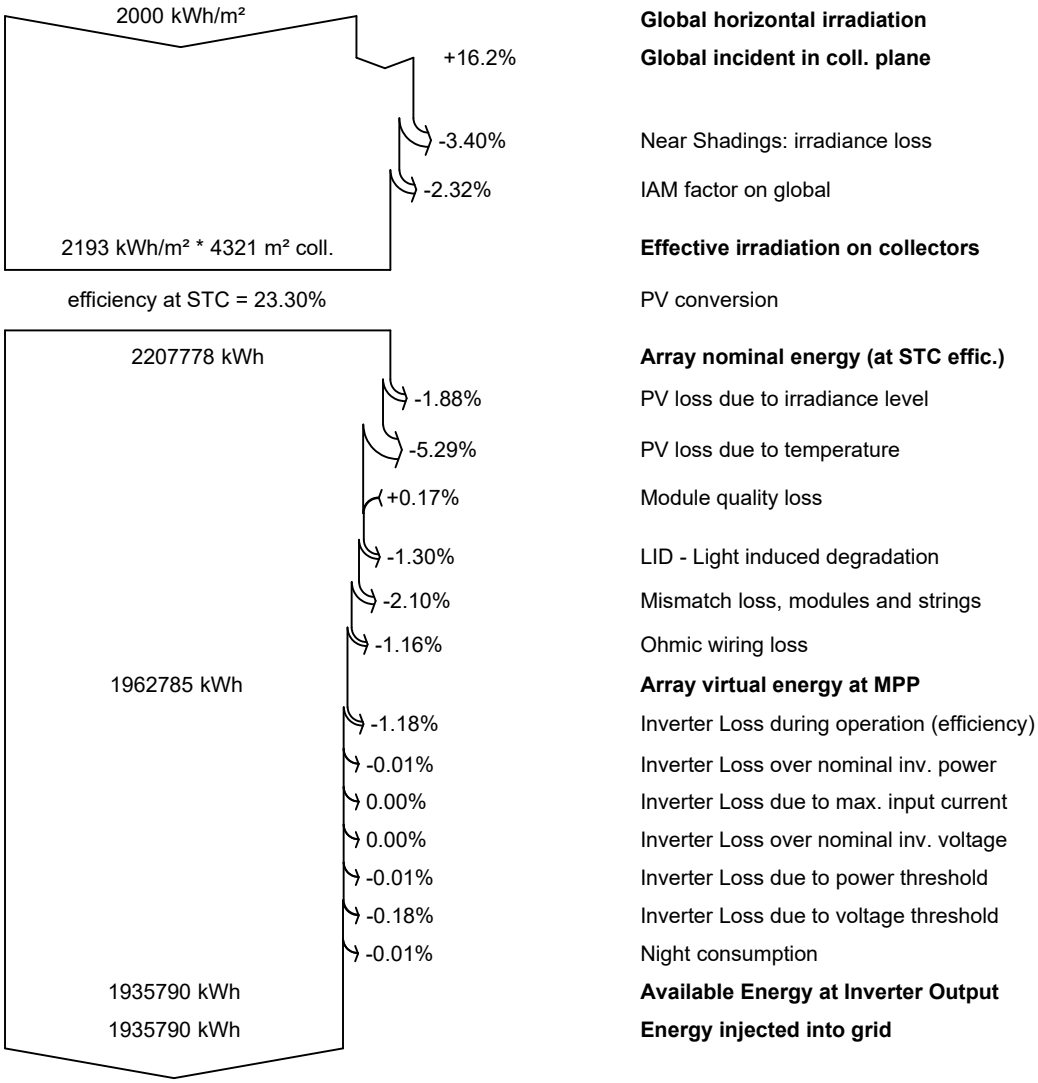
	GlobHor kWh/m²	DiffHor kWh/m²	T_Amb °C	GlobInc kWh/m²	GlobEff kWh/m²	EArray kWh	E_Grid kWh	PR ratio
January	102.0	26.77	5.11	170.0	155.3	146971	145133	0.852
February	116.1	33.22	7.98	167.1	159.6	149343	147505	0.881
March	149.8	57.49	13.89	180.4	172.1	157648	155748	0.861
April	178.7	66.78	18.94	187.9	178.0	160277	158380	0.841
May	214.2	73.20	25.50	204.0	192.8	169645	167664	0.820
June	238.8	68.10	30.84	214.7	202.9	174623	172613	0.802
July	241.3	67.65	34.12	222.7	211.0	178676	176614	0.791
August	226.1	54.26	32.31	229.9	219.2	186981	184786	0.802
September	193.6	35.38	27.01	227.2	217.9	189932	187666	0.824
October	144.9	37.72	20.32	197.1	189.2	169315	167289	0.847
November	102.2	29.63	11.28	162.3	150.2	139114	137433	0.845
December	92.5	23.77	6.47	160.7	144.8	136623	134960	0.838
Year	2000.4	573.97	19.55	2324.0	2193.0	1959148	1935790	0.831

Legends

GlobHor	Global horizontal irradiation
DiffHor	Horizontal diffuse irradiation
T_Amb	Ambient Temperature
GlobInc	Global incident in coll. plane
GlobEff	Effective Global, corr. for IAM and shadings
EArray	Effective energy at the output of the array
E_Grid	Energy injected into grid
PR	Performance Ratio

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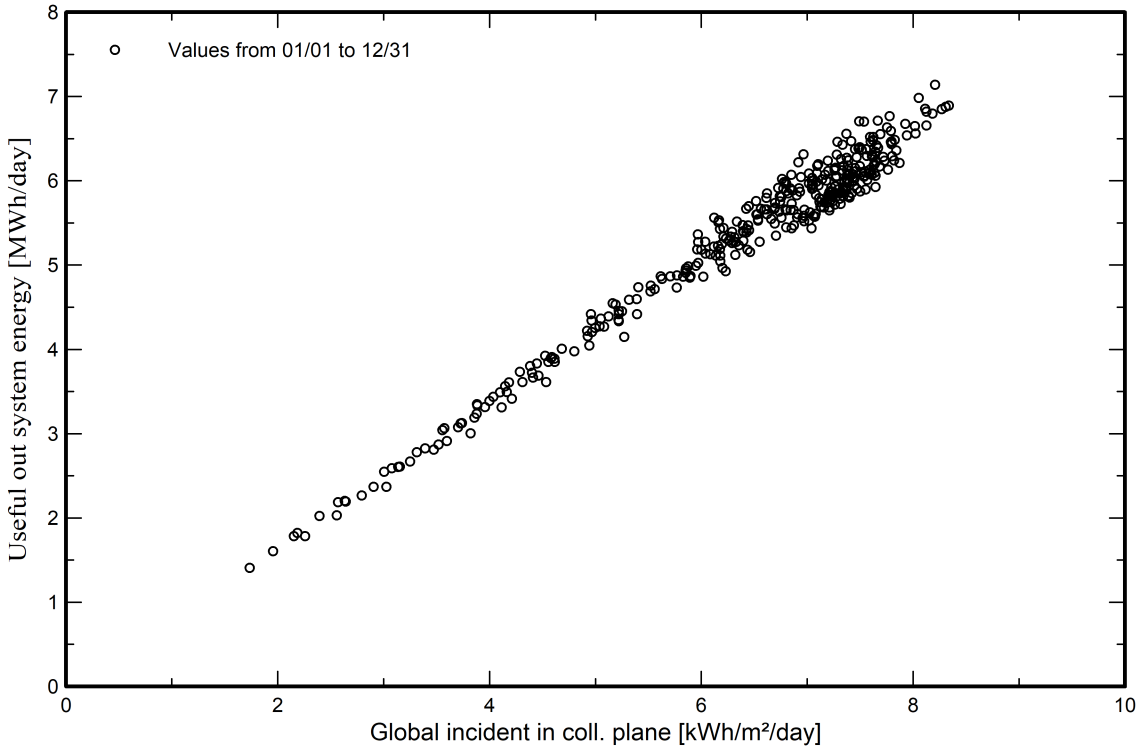
Loss diagram



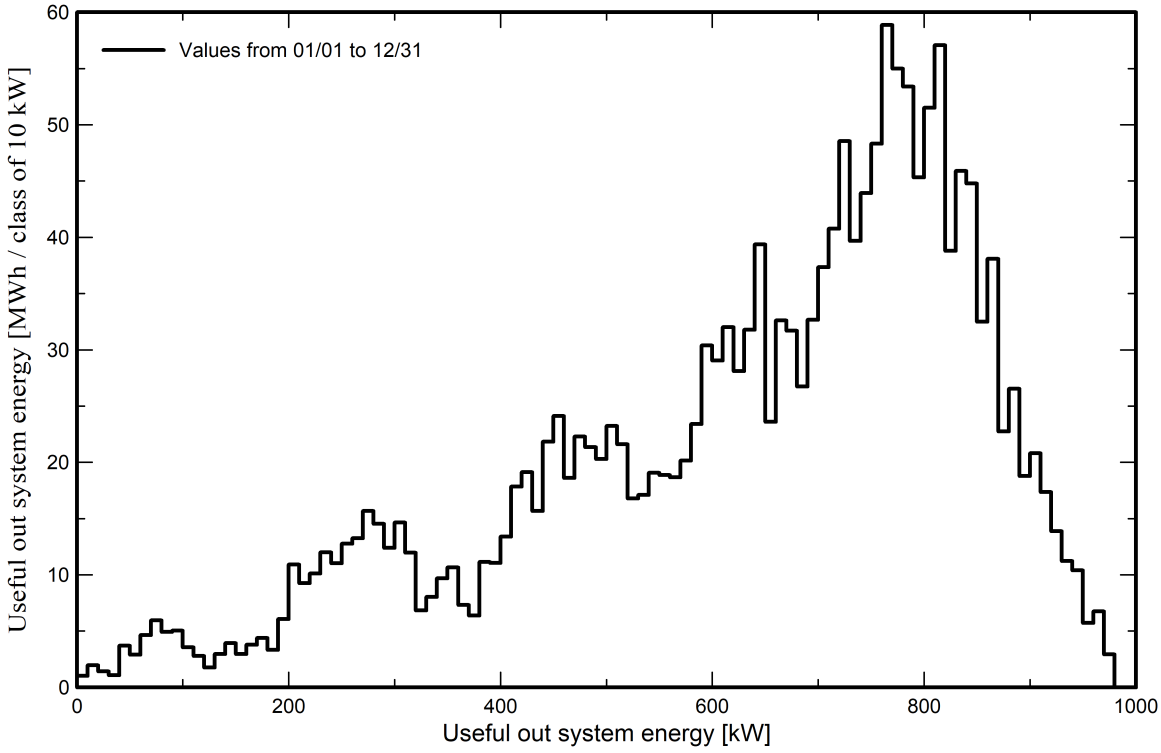
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Predef. graphs

Daily Input/Output diagram

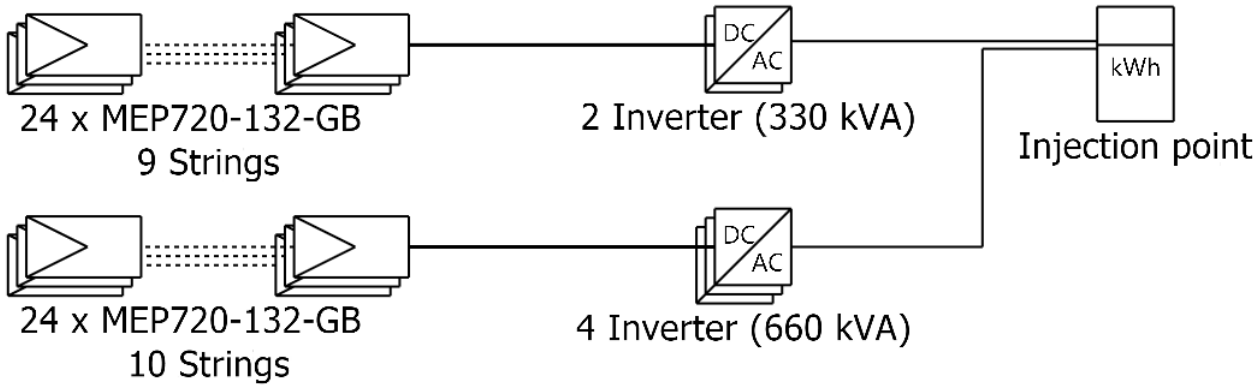


System Output Power Distribution



Single-line diagram

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PV module	MEP720-132-GB
Inverter	Sunny Highpower SHP165-US-21-PEAK3
String	24 x MEP720-132-GB

	Qom-1MW	
	VC0 : New simulation variant	10/25/26